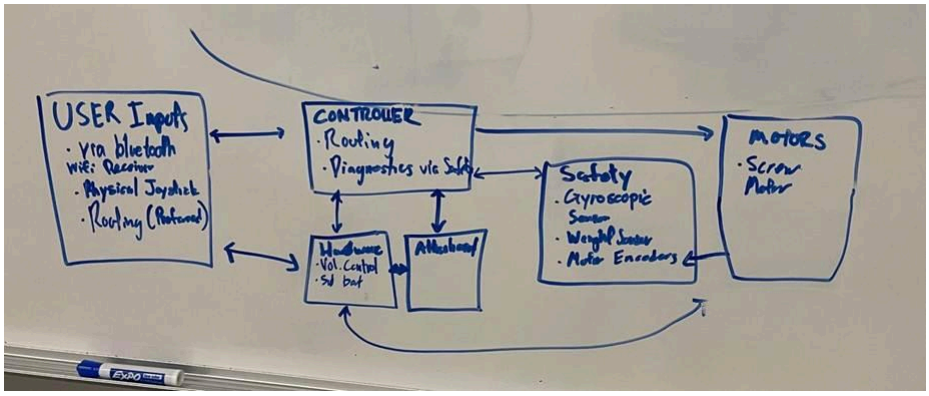
Gantry - Design Feedback



Quick reminder

- A box is one part of the system with one job. It is either where a person interacts (an app or screen), where the work happens (server or controller), or where things are stored (database).
- An arrow shows something moving from one box to another. Write on the arrow what is moving. The arrow points to where the data goes.
- If a box has nothing coming into it, think about how it gets its information.

Comments on your diagram

1. Most arrows go both ways and have no labels. Write what moves each way, for example "target position" one way and "current position" the other.
2. The Safety box lists sensors. Sensors are parts that send readings. Show them as a box that sends to the controller, and show which box makes the safety decision.
3. There is no emergency stop. It should cut power to the motors directly, so it still works if the controller freezes.
4. User Inputs lists Bluetooth, Wi-Fi, a joystick, and routing. Pick one for Version 0.
5. It is not clear what moves between Hardware, Attachment, and the other boxes.

Where to start

Start where the target comes in (the joystick or a typed command), and follow it until the hook moves. Then follow a sensor reading back to the controller.

Try this example

Walk this story through your diagram, one step at a time. For each step, write which box does the work and what moves on the arrow, then write what is missing from your diagram (a box, an arrow, a label, or nothing).

Step	Which box does the work?	What moves on the arrow?	What's missing?
1. The gantry powers on. It does not know where the hook is yet.	User input, sensors, controller	User Inputs and Readings	nothing
2. The operator sends the target (40, 25) before homing. It is refused.	User Input, Controller	Movements, System Status	nothing
3. After homing, the operator sends the same target. The hook moves there.	User Input, Controller, Motor	User input, Action, Motor Status	nothing
4. The hook picks up a 3 kg box. The weight is checked before it moves.	Sensor, Controller	Weight	nothing
5. Halfway through a move, the controller freezes. The motors stop anyway.	Controller, Motor	Action	nothing
6. Power comes back. The gantry finds its position again before it moves.	Controller, User input	System Status	nothing